Network Partitions

both sides alive, neither can reach the other

Watch the 4-min video

In one line: a network partition splits nodes into groups that are all running but can't talk to each other; during one you must choose between consistency (only a majority side serves writes) and availability (every side serves and copies diverge), and you should decide that before it happens.

1 What a partition is

Two data centers run one database cluster. The link between them fails. Both sides are healthy and both can still serve their users, but they can't reach each other. That's a **network partition**: the cluster is split into groups that can only talk within themselves. Causes are ordinary: a cut fibre, a misconfigured switch or firewall rule, a cloud networking incident, or a node so overloaded or paused (GC) that it looks unreachable.

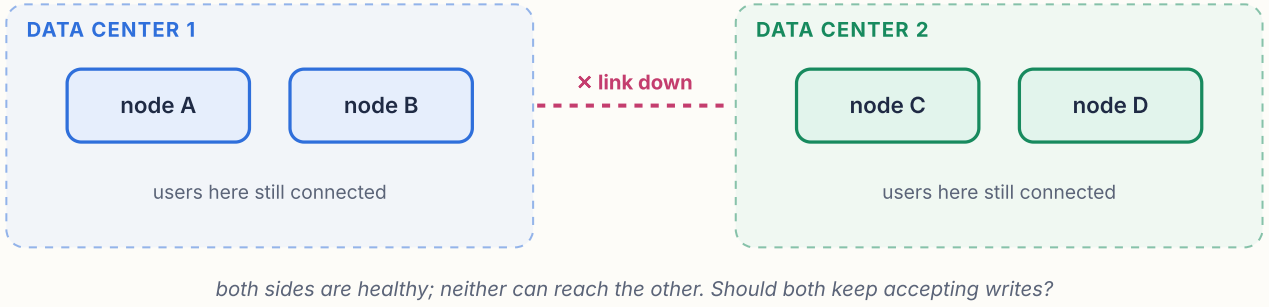


Figure 1. A partition: every node is up, but the link between the two groups is gone.

2 From inside, a partition looks like failure

Node A stops hearing from node C. Did C crash, or is the path to C broken? From A's side it's impossible to tell. Both cases look like silence. So systems must make the same safe choice either way, usually by counting how many nodes they can still reach.

3 The choice: consistency or availability

The **CAP theorem** says that while a partition lasts, a system that replicates data can't be both **consistent** (every read sees the latest write) and **available** (every non-failed node answers). If both sides accept writes to the same data, the copies diverge. If only one side may write, the other side must refuse.

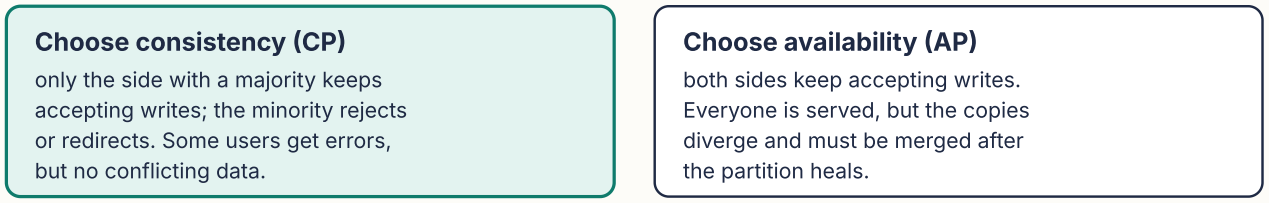


Figure 2. The two options during a partition. Banks and lock services choose CP; shopping carts and like counters often choose AP.

CP systems (etcd, ZooKeeper, Spanner, most leader-based databases) let only the side holding a **majority** (a quorum) keep the leader. **AP systems** (Dynamo-style stores like Cassandra at low consistency levels) keep both sides writable and reconcile later with last-write-wins, vector clocks or CRDTs (lessons 7 and 19).

WORKED EXAMPLE: AN AVAILABILITY-ZONE LINK FAILS

A link failure separates the database primary in zone A from its replicas in zones B and C. B and C can still see each other: together they are a majority of three, so they elect B as the new primary and the app in B and C keeps writing. The old primary in zone A is alone, a minority, so it must **step down** and reject writes. If it kept accepting writes from clients in zone A, there would be two primaries and conflicting data: split-brain (lesson 12). When the link heals, A rejoins as a replica of B.

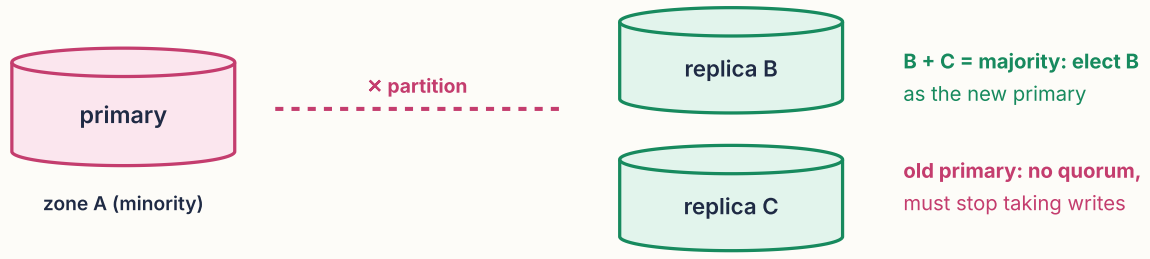


Figure 3. The majority side elects a new primary; the isolated minority must stop accepting writes.

4 Healing and testing

When the partition heals, AP systems must **merge** divergent writes, and CP systems must bring the minority back up to date. Partitions also happen partially (A sees B, B sees C, A can't see C) and asymmetrically (messages flow one way). Test them deliberately with tools that drop traffic between nodes, as Jepsen-style testing does.

COMMON MISTAKES

- Assuming partitions are too rare to plan for.
- Two-node clusters: neither side can ever have a majority.
- Letting an isolated primary keep taking writes because it 'is still healthy'.
- Choosing AP without a plan to merge conflicting writes.

5 Key terms

Network partition: nodes are up but split into groups that can't communicate

Quorum / majority: more than half of the nodes

CP: stays consistent during a partition, refusing some requests

AP: stays available during a partition, allowing divergence

Split-brain: two sides each believe they are the leader

Heal: the partition ends and nodes reconnect

6 Interview questions

Q1. What is a network partition? BASIC

A failure where nodes keep running but some groups can't reach each other, so the cluster is split. It looks the same as remote nodes crashing.

Q2. What does CAP say? BASIC

During a partition, a replicated system must choose: stay consistent by refusing some requests, or stay available and let copies diverge. Outside partitions, it can provide both.

Q3. Your primary is cut off from its two replicas. What should happen? SCENARIO

The replicas, a majority, elect a new primary. The isolated primary, with no quorum, must stop accepting writes; fencing makes sure its writes are rejected if it tries.

Q4. A shopping cart service and a bank ledger: CP or AP? SCENARIO

The cart can be AP: accept adds on both sides and merge them later. The ledger should be CP: refusing a transfer briefly is better than double-spending.

Q5. Why do clusters use an odd number of nodes? TRICKY

To make a majority possible on one side of a split. With 3 or 5 nodes, one side always has more than half; with 2 or 4, an even split leaves no majority, so nothing can write.

Q6. Can a node tell a partition from a crash? TRICKY

No. Both look like missing messages. That's why decisions are based on quorums and leases, not on guessing what happened to the silent node.

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